Introduction to Simulink and Stateflow

Lab 1, ESDM

Objective

Introducing students to the Simulink environment and the Stateflow component.

Theoretical aspects

TBD

Exercises

1. Create a Parking Counter FSM according to the specifications below.

   Steps:

   - Create a new Simulink model
   - Create a Stateflow Chart inside the model
   - Design the FSM inside the chart
   - Add some inputs and test the model

   Model specifications

   Inputs:

   - up: boolean. Whenever a car is entering the area, `up` is TRUE.
   - down: boolean. Whenever a car is leaving the area, `down` is TRUE.

   Outputs & internal variable:

   - count: integer [0,10]

Requirements:

- The output `count` shall indicate always the current number of cars in the parking area.
- Upon initialization, `count` shall be initialized to 0.

2. Extend the model in the following way:

   - The inputs `up` and `down` shall be considered upon transitioning from FALSE to TRUE (e.g. if `up` stays TRUE for 10 seconds, count it only once).
   - Introduce another input `count_init` which shall be used as the initialization value for `count`.

General requirements

- Model Settings: Set the Solver type to "Fixed-step", "discrete (no continuous states")", and fixed step size to 0.1 (see Fig.1)

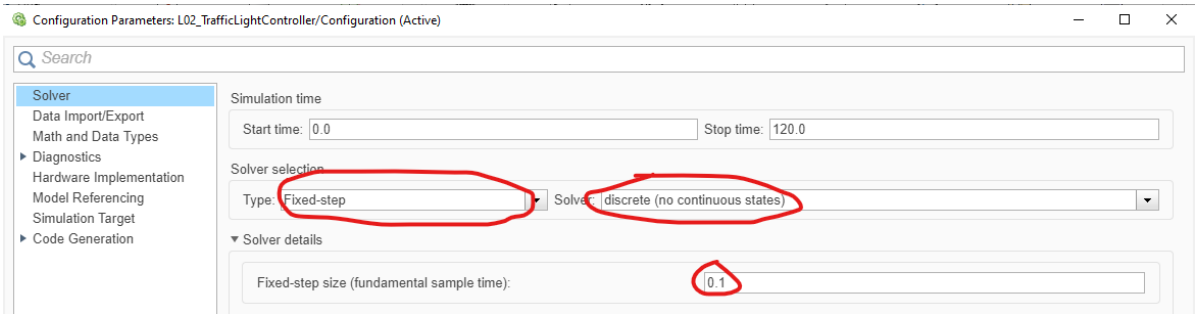


Figure 1: Model Settings

Final questions

1. TBD